

construct portfolios with securities weighted by various metrics of fundamental value and offer a way to collect a value premium.²⁹ These exposures are not determined directly by the investor but are a first step in collecting dynamic risk premiums. True factor investing calls for an individual-specific benchmark, and there has been an increasing trend to customization in the index business.

4.2. GM ASSET MANAGEMENT

The dynamic factors involve trading, and thus minimizing costs and turnover, are essential components in constructing factor benchmarks. The largest investors have constructed and trade their own factor portfolios.

GM Asset Management, the fund manager of the parent company's pension plan, has built in-house dynamic factor portfolios for value and momentum.³⁰ James Scott (2012), the managing director of GM Asset Management's global public markets business, says that their approach is "consistent with the management of alpha and risk at the plan level and surplus plan management as suggested by Ang, Goetzmann and Schaefer (2009)."

GM Asset Management starts with an institutionally-investible universe. They do not short any growth stocks or loser stocks—they either underweight them relative to market weight or hold them at a zero weight. Their measure of fundamental value takes an average of Book/Price, Forward Earnings/Price, Price/Sales, Trailing Earnings/ Price, and Trailing Cash Flow/Price. They define momentum as the past twelve month return (but drop the most recent month, following standard academic practice, to avoid a reversal effect). Portfolios are rebalanced monthly and designed for low turnover. The factor portfolios have only small exposures to value and momentum, reflecting GM Asset Management's desire for only small factor exposures (they have tight *tracking error*). They have built U.S. value and momentum portfolios, as well as international versions.

GM Asset Management benchmarks all its equity managers to these in-house factor portfolios. Managers have to prove they are generating excess returns beyond those available passively from the factor benchmarks. They assess managers by a regression:

$$r_{active} = r - r_{mkt} = b_0 + b_1 r_{mkt} + b_2 r_{size} + b_3 r_{value-growth} + b_4 r_{momentum} + \varepsilon, \quad (14.6)$$

where r_{mkt} is a standard market-weighted aggregate benchmark, r_{size} proxies for the size effect (they take the Russell 2000 minus the Russell 1000), and

²⁹ See comments by Perold (2007), Jun and Malkiel (2008), and West (2010).

³⁰ For further details see "GM Asset Management and Martingale's Low Volatility Strategy," Columbia CaseWorks ID #110315. This section reflects conversations with Scott.